

CH1101: Quantum Chemistry and Spectroscopy

Tutorial Sheet 01

Module: Quantum Chemistry

- Q1. The regions of electromagnetic radiation are usually described in terms of wavelength λ , and are given below in meters (m). Calculate the values of $\tilde{\nu}$, ν , and E for those regions. Compare your results on radiations of various ranges: (i) Energy vs. wavelength, (ii) Energy vs. frequency

Wavelength, λ [m]	10^{-10}	10^{-8}	10^{-6}	10^{-4}	10^{-2}	1
Wavenumber, $\tilde{\nu}$ [cm^{-1}]						
Frequency, ν [Hz]						
Energy, E [$Jmol^{-1}$]						
	$X\text{-rays}$		$UV\text{-Visible}$	$Infrared$	$Microwaves$	$Radiowaves$

[Note: The energy of a photon is given by Planck's law

$$E = h\nu = h\frac{c}{\lambda} = hc\tilde{\nu}$$

where, h is the Planck constant, c is the speed of light.]

Table 1: Useful physical constants and conversion factors

Speed of light (c)	=	$2.997 \times 10^8 \text{ m s}^{-1}$
Mass of electron (m_e)	=	$9.109 \times 10^{-31} \text{ kg}$
Planck constant (h)	=	$6.626 \times 10^{-34} \text{ J s}$
Boltzmann constant (k_B)	=	$1.381 \times 10^{-23} \text{ J K}^{-1}$
Avogadro number (N_A)	=	$6.023 \times 10^{23} \text{ mol}^{-1}$
Elementary charge (e)	=	$1.602 \times 10^{-19} \text{ C}$
Mass of proton (m_p)	=	$1.672 \times 10^{-27} \text{ kg}$
Rydberg Constant (R_H)	=	109677 cm^{-1}
1 eV	=	$1.602 \times 10^{-19} \text{ J}$